Wednesday Lab

Q1 Write a program that takes n students (name and
roll number) as input using structures and then sorts the student list based on
their roll numbers and display that.
Test Data :
Input the string :
Enter number of student record to insert: 4
Enter name for student 1: ram
Enter roll number for student 1: 12
Enter name for student 2: shayam
Enter roll number for student 2: 5
Enter name for student 3: love
Enter roll number for student 3: 78
Enter name for student 4: kush
Enter roll number for student 4: 2
Expected Output :
Sorted list of students based on roll number :
Name: kush, Roll Number: 2
Name: shayam, Roll Number: 5
Name: ram, Roll Number: 12
Name: love, Roll Number: 78

Q2.Write a C program to create a singly linked list and insert a new node at a specified position. The program should take the number of nodes and their data as input, then take the position and value for the new node. Finally, display the linked list.
Example : Input the number of nodes: 3
Input data for node 1: 10
Input data for node 2: 20
Input data for node 3: 30
Input the position to insert the new node: 2
Input data for the new node: 15
Output: The updated linked list is: 10 -> 15 -> 20 -> 30

Q3 Write a C program to create a singly linked list and delete a node from the beginning of the list. The program should take the number of nodes and their data as input, and then delete the node from the beginning as many times as the user wants. Finally, display the updated linked list.
ExampleInput the number of nodes: 4
Input data for node 1: 10
Input data for node 2: 20
Input data for node 3: 30
Input data for node 4: 40
Do you want to delete a node from the beginning? (1 for yes, 0 for no): 1
Do you want to delete a node from the beginning? (1 for yes, 0 for no): 1

Do you want to delete a node from the beginning? (1 for yes, 0 for no): 0

The updated linked list is: 30 -> 40

Q4 Write a C program to create a file called example.txt, write the sentence "Hello, File Handling!" into it, and then read and display the content of the file.
Example:
Content of the file example.txt:
Hello, File Handling!

Q5. Write a C program to create a file named `data.txt`. The program should take a user's name and age as input, store them in the file, and then read and display the content of the file.

Input your name: John
Input your age: 25

The content of the file data.txt is:
Name: John, Age: 25